

CSCC63 --- Computability & computational complexity

Week 2 Tutorial

Practice with Turing machines;
decidable and recognizable sets

Question 1 (cont'd)

Design a Turing machine that computes the "increment-by-one" function:

Given a number in binary on the tape, increment the number by one and halt.

You can assume that the input string is not empty (the empty string does not represent any number in binary).

What should the output be if the input is 10010111?

10011000

What should the output be if the input is 11111111?

100000000

Question 1 (cont'd)

High level description:

1. Scan right to the first (leftmost) blank, remembering if you saw a 0 (the TM "remembers" things in its state).
2. If in Step 1 a 0 was found, scan left until the first (rightmost) 0, changing every 1 encountered to 0 and the first 0 to 1, and halt.
3. If in Step 1 no 0 was found, replace the leftmost blank by a 0, and replace all the 1s by 0s, except the leftmost one (the one in cell 1).

Question 1

How can we tell if we are in cell 1?

Various ways, e.g.:

- replace the symbol on the current cell by a special one, say #, remembering (in the state) what the old symbol was, and move left
- if the symbol now under the tape head is # then this is cell 1 (moving left didn't change the position of the head); replace # by the symbol that was there
- if not, go right without changing the symbol, change the # to the old symbol and move left, continuing the search for cell 1.

Question 1

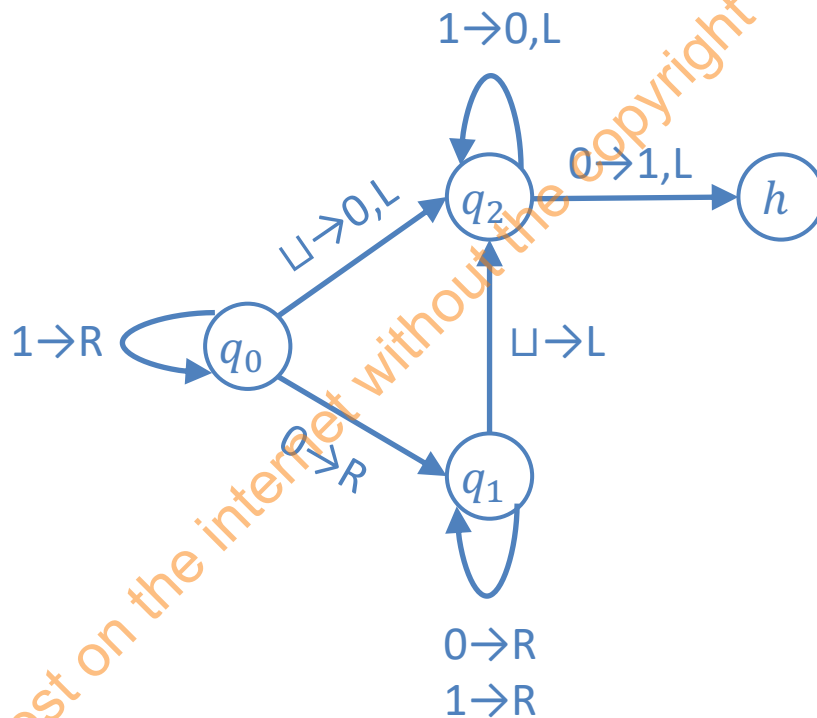
In our case we can exploit the fact that we know that all symbols encountered in Step 1 were 1s:

If in Step 3 (i.e., the case when in Step 1 we found no 0, and we are scanning left changing every 1 to a 0) we find that the tape head is scanning a 0, we must be in cell 1: the last move to the left didn't change the position of the tape head and we are seeing the 0 we just wrote!

So, we can change that 0 to a 1, and halt.

Question 1

Diagram for transition function of this TM:



Question 2

The set of decidable languages is closed under intersection.

True or false?

True!

Let M and M' be TMs to decide languages L and L' .

TM to decide $L \cap L'$:

on input x :

run M on x

run M' on x

if both accept, then accept
else reject

What does "run M on x " mean?
Incorporate the transition function of M into the transition function of the TM to decide $L \cap L'$ that we are constructing to see if $x \in L$.
Similarly for "run M' on x ".

Question 2 (cont'd)

The set of recognizable languages is closed under intersection.

True or false?

True!

Let M and M' be TMs to recognize languages L and L' .

TM to recognize $L \cap L'$:

on input x :

run M on x

run M' on x

if both accept, then accept

else reject

What if M or M' loops on x ?

No problem: Then $x \notin L \cap L'$.

Looking for recognizer, not a decider, so this is OK!

Question 3

The set of decidable languages is closed under union.

True or false?

True!

Let M and M' be TMs to decide languages L and L' .

TM to decide $L \cup L'$:

on input x :

run M on x

run M' on x

if at least one accepts, then accept

else reject

Identical to intersection, except
replace “both” by “at least one”

Question 3 (cont'd)

The set of recognizable languages is closed under union.

True or false?

What about: same as before (as we did in intersection)...

Let M and M' be TMs to recognize languages L and L' .

TM to recognize $L \cup L'$:

on input x :

run M on x

run M' on x

if at least one accepts, then accept

else reject

What if M loops on x but $x \in L'$?

This does not work ☹️

Question 3 (cont'd)

So...

is the set of recognizable languages closed under union?

Yes

To recognize $L \cup L'$:

on input x :

for each $k := 1, 2, \dots$ do

run M on x for k steps

run M' on x for k steps

if at least one accepts, then accept

else go on to the next k

On your own: Why does this work?

Question 3 (cont'd)

"Loops" in TMs.

How do we do "for each $k := 1, 2, 3, \dots$ " with TMs?

- Use an additional tape in which to store k , initially 0.
- At the start of each iteration, increment the value stored on that tape (using the TM in Question 1); use that value when k is referenced.
- For example, to do something k times (as in "run M on x for k steps"), start a counter at 0 in another tape, and increment it by one each time, until it becomes k . Check this by comparing bit-by-bit the counter with the tape that contains k .

Question 4

In Turing's original definition the Turing machine tape extended to infinity in both directions.

This is called a two-way infinite tape TM (TTM).

- the input is placed somewhere on the tape
- all other cells contain blanks
- the tape head is positioned on the cell containing the leftmost symbol of the input
- the machine takes steps as a regular TM
- no “moving left from the leftmost cell” problem.

Question 4 (cont'd)

Show that TMs and TTMs are equivalent: each can simulate the other.

We say that M simulates M' if, for every input x ,

- M' accepts $x \Rightarrow M$ accepts x
- M' rejects $x \Rightarrow M$ rejects x
- M' loops on $x \Rightarrow M$ loops on x

Need to show how to:

- 1) Simulate a TM M using a TTM M'
- 2) Simulate a TTM M using a TM M'

Question 4 (cont'd)

(1) is easy: Simulate a TM M using a TTM M' .

- Place special marker $\#$ in the cell to the left of input
- Follow the transition function of the TM
- If head is ever on cell containing $\#$, move R

Easy to see that

M' accepts/rejects/loops on x

$\Leftrightarrow M$ accepts/rejects/loops on x .

Question 4 (cont'd)

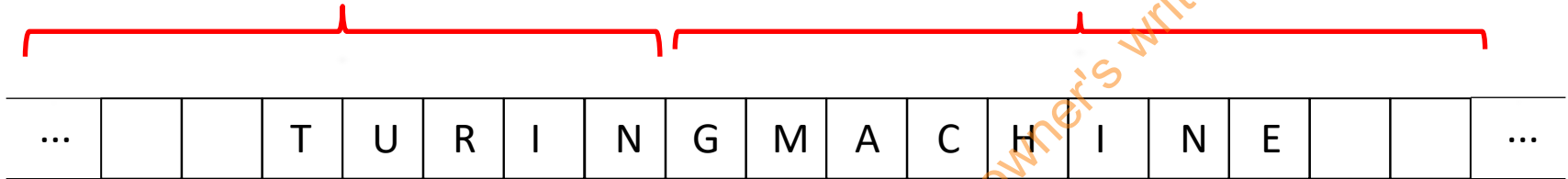
(2) is more interesting: Simulate a TTM M using a TM M'

Idea #1: Use a two-track one-way infinite tape to store the two-way infinite tape of M "folded" in half at the cell where the input started. Put a # on the leftmost cell.

Question 4 (cont'd)

— part of the tape

+ part of the tape



2-way infinite tape of TTM M

cell initially holding
leftmost input symbol

#	G	M	A	C	H	I	N	E					...
	N	I	R	U	T								

1-way infinite tape of simulating TM M'

Question 4 (cont'd)

- upper track contains symbols in cells $0, 1, 2, \dots$ of TTM (+ part of its tape)
- lower track contains symbols in cells $-1, -2, \dots$ of TTM (– part of its tape)
- leftmost cell of TM contains special symbol #
- the state of TM keeps track of
 - the state of TTM
 - whether the simulated head of TTM is in + or – part of the tape, and
 - whether the last simulated move of TTM was L or R.

Question 4 (cont'd)

- Initially write # to the leftmost cell, and shift the input to the R by one cell, copying the input symbol on the upper track and putting a blank on the lower track (i.e., changing a to $(a, \sqcup)$).
- Move head to the cell that is to the R of # and start taking steps.
- While in + part, follow the TTM transition function, based on (and changing) the upper track symbol.
- While in – part, follow TTM transition function, based on (and changing) the lower track symbol, but reverse direction: if the TTM moves R, make the TM move L, and vice-versa.

Question 4 (cont'd)

- Moving from the $+$ to the $-$ part of the TTM's tape:
If the head is on the symbol $\#$, and the state indicates that
 - a) the (simulated) tape head of the TTM is in the $+$ part of its tape, and
 - b) the last move was L,change the state of the TM to indicate that we are now in the $-$ part of simulated TTM's tape.
- Similarly for moving from the $-$ to the $+$ part of the TTM's tape.

Question 4 (cont'd)

(2) Simulate a TTM M using a TM M' (alternative 2)

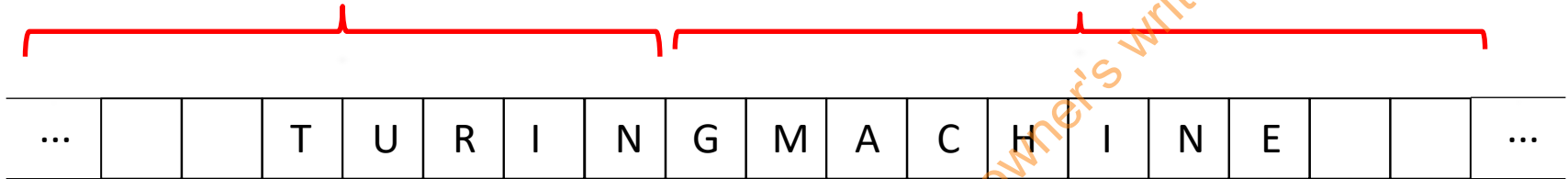
Idea #2: Similar to Idea #1, but use two tapes instead of a two-track single tape machine.

Since a single-tape TM can simulate a two-tape TM, we can simulate the TTM using an ordinary single-tape TM.

Question 4 (cont'd)

— part of the tape

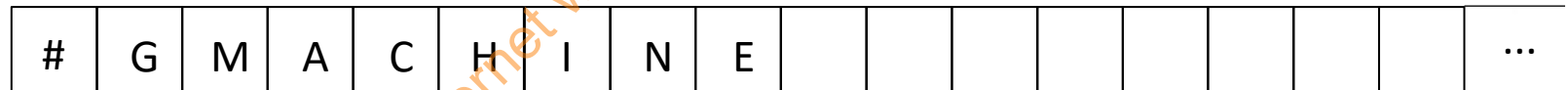
+ part of the tape



2-way infinite tape of TTM M

cell initially holding
leftmost input symbol

tape 1



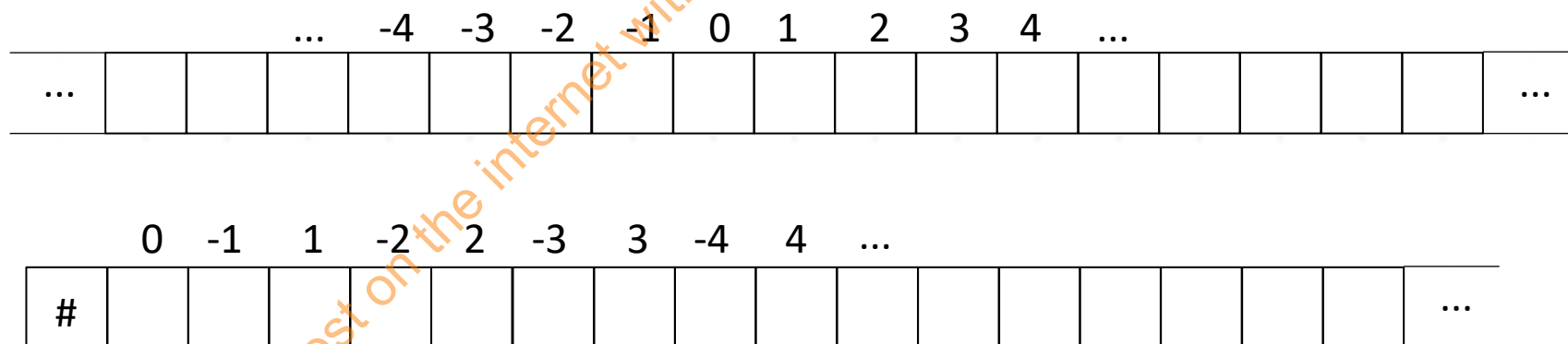
tape 2



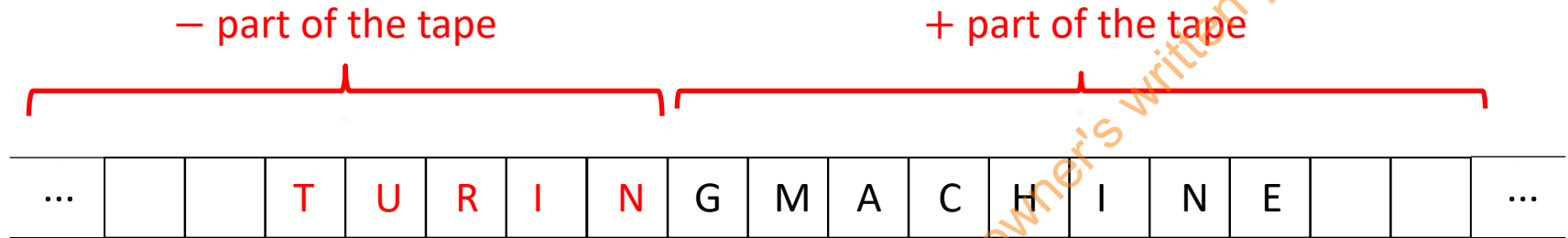
Question 4 (cont'd)

(2) Simulate a TTM M using a TM M' (alternative 3)

Idea #3: Dovetail the + and – parts of the TTM's tape.



Question 4 (cont'd)



2-way infinite tape of TTM M

cell initially holding
leftmost input symbol

#	G	N	M	I	A	R	C	U	H	T	I		N		E		...
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tape of simulating TM M'

On your own: Work out ideas #2 and #3 in more detail